AEGIS FORENSICS

An Integrated Artificial Intelligence & Digital Forensic Framework for Email Threat Detection and Deepfake Media Analysis

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EXECUTIVE DOSSIER PURPOSE: *This master document contains an exhaustive technical, mathematical, and legal breakdown of the AEGIS Forensics system. It is authored to provide examiners, faculty evaluators, and technical interviewers with definitive insight into system genesis, decoupled architecture, machine learning models, computer vision pipelines, frequency domain transforms, binary stream introspection, court admissibility protocols, and viva voce defense readiness.*

1. System Genesis, Core Purpose & Architectural Decoupling

Modern digital investigations face two acute cyber threats: **adversarial spear-phishing** (accounting for over 91% of enterprise breaches according to CISA) and **generative deepfakes** (synthesized via GANs and Latent Diffusion models). Historically, investigators have been hindered by disparate command-line scripts, black-box AI predictions that collapse under judicial scrutiny, and chain-of-custody violations that render evidence inadmissible in legal proceedings.

The Principle of Operational Decoupling: AEGIS Forensics implements two strictly decoupled investigative modules: **Module A (Email Threat Forensics)** and **Module B (Deepfake Image Forensics)**. While both modules share the unified web dashboard and SQLite persistence layer, their processing pipelines never cross-contaminate. In digital forensic jurisprudence (ISO/IEC 27037), evidence separation ensures that textual anomalies cannot artificially skew visual manipulation findings, guaranteeing that either module can be admitted independently in a court of law.

Subsystem	Technology Stack	Forensic Role & Key Responsibilities
Backend Gateway	Python 3.10+, FastAPI, Uvicorn (ASGI)	High-concurrency asynchronous REST API, request schema validation (Pydantic v2), route orchestration.
Email ML Engine	Scikit-Learn, Joblib, TF-IDF, Logistic Regression	Sublinear TF-IDF vectorization, calibrated logistic regression (98.20% accuracy), token classification.
Explainable AI (Text)	LIME (Local Interpretable Explanations)	Local token perturbation, exponential distance kernel, visual feature attribution bars.
Deepfake Vision CNN	PyTorch, Torchvision, ResNet-18	Spatial-residual feature extraction, 512-dim bottleneck, binary deepfake/authentic classification head.
Visual Explainability	Grad-CAM (Layer-4 Activation Maps)	Global average pooling of gradients, ReLU-weighted activation heatmap overlay via OpenCV JET.
Frequency Analysis	NumPy, SciPy.fftpack, OpenCV	2D Fast Fourier Transform (log-magnitude), 2D Discrete Cosine Transform (energy decay), spectral centroid.
Binary Stream Inspector	Pillow, Custom Bitstream Parsers	JPEG SOF0 chroma subsampling (4:2:0/4:2:2/4:4:4), DQT quantization quality estimation, EXIF recovery.
Court PDF Engine	ReportLab 5.x (Two-Pass Canvas)	12-section court-admissible PDF generator, dual-hash verification, dynamic custodian, 28 U.S.C. § 1746 oath.
Frontend Cockpit	React 18, TypeScript, Vite 6, Tailwind v4	Reactive cyber-forensic interface, dark/light theme, real-time metrics, custodian input card, exhibit viewer.

2. Deep-Dive: Module A — Email Threat Forensics Pipeline

2.1 Mathematical TF-IDF Representation & Logistic Classifier

Incoming emails are parsed through an RFC 5322 header and body extractor. The text stream is converted into a mathematical feature vector using Term Frequency-Inverse Document Frequency (TF-IDF) with sublinear term-frequency scaling and n-gram ranges (1, 2):

$$\text{TF-IDF}(t, d, D) = (1 + \ln(\text{TF}(t, d))) \times \left[\ln\left(\frac{1 + |D|}{1 + |\{d \in D : t \in d\}|}\right) + 1\right]$$

The resulting high-dimensional sparse vector is evaluated by a calibrated Logistic Regression classifier with L2 regularization ($C=1.0$):

$$P(\text{Phishing} \mid \mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + b)}}$$

Empirical Validation: Trained on an enterprise corpus of 18,650 labeled emails, the model achieved **98.20% accuracy**, **0.9762 F1-score**, **97.80% precision**, and **97.45% recall**, with an average inference latency of ~42 milliseconds.

2.2 Local Interpretable Model-agnostic Explanations (LIME)

To comply with legal transparency standards and overcome the 'black-box' defense, AEGIS incorporates LIME. For an email instance x, LIME generates $N=500$ perturbations $z' \in \{0, 1\}^d$ by randomly zeroing out words, evaluates predictions $f(z)$, weights instances using an exponential kernel $\pi_x(z) = \exp(-D(x, z)^2 / \sigma^2)$, and solves a weighted ridge regression to find token importance weights w_g. Tokens such as *'urgent'*, *'verify'*, *'suspend'*, and *'immediately'* are highlighted with exact numerical risk contributions.

2.3 Lexical URL & Authentication Header Engines

• **Authentication Headers:** Inspects RFC 7208 (SPF), RFC 6376 (DKIM), and RFC 7489 (DMARC) headers to detect sender forgery. Crucially, it verifies the Reply-To header against the display From address to catch deceptive redirection.
• **Lexical URL Engine:** Extracts fully qualified domains via tldextract. It flags raw IP hosts (e.g. http://192.168.1.1/login), high-risk TLDs (.xyz, .top, .tk), and brand spoofing using Levenshtein distance matching (e.g. paypa1.com vs paypal.com).
• **Attachment Hazard Scoring:** Evaluates extensions against known malware vectors (.exe, .vbs, .scr, .iso).
• **Composite Multi-Factor Score:** Synthesizes ML probability (45%), URL threats (30%), writing urgency (15%), and attachment risks (10%) into a unified 0–100 score.

3. Deep-Dive: Module B — Deepfake Image Forensics Pipeline

3.1 ResNet-18 Spatial-Residual Neural Architecture

AEGIS employs an optimized ResNet-18 backbone. Unlike standard classifiers that seek semantic object labels, forensic classification seeks subtle high-frequency interpolation errors, boundary blurring, and textural inconsistencies left by convolutional upsampling layers.

• **Input:** Normalized 3-channel tensor ($3 \times 224 \times 224$) extracted via OpenCV Haar face detection (or aspect-ratio center crop).
• **Residual Skip Connections:** Solves vanishing gradient degradation via identity shortcuts: $\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x}$, enabling the network to propagate spatial gradient residuals directly through 4 residual stages (64, 128, 256, and 512 channels).
• **Forensic Classification Head:** Replaces standard 1000-class ImageNet head with AdaptiveAvgPool2d(1x1) $\rightarrow$ Dropout(p=0.3) $\rightarrow$ Linear(512, 2) with Softmax yielding calibrated probabilities for [Authentic, Deepfake].

3.2 Layer-4 Gradient-Weighted Class Activation Mapping (Grad-CAM)

To establish visual explainability under the Daubert test, AEGIS taps the final convolutional layer (Layer 4) of ResNet-18. It captures gradients of the target class score Y^c with respect to feature activation maps A^k ($512 \times 7 \times 7$):

$$\alpha_k^c = \frac{1}{Z} \sum_{i=1}^7 \sum_{j=1}^7 \frac{\partial Y^c}{\partial A_{i,j}^k}, \quad L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_{k=1}^{512} \alpha_k^c A^k\right)$$

The resulting coarse 2D heatmap is upsampled to input resolution, normalized between [0, 1], mapped to false colors via OpenCV COLORMAP_JET, and overlaid on the suspect image at 50% opacity. This reveals whether the model focused on authentic anatomical features or synthetic artifacts around eyes, teeth, and blended face boundaries.

3.3 Dual-Domain Frequency Spectrum Analysis (2D FFT & 2D DCT)

Generative models (GANs and Diffusion models) leave distinct periodic spectral artifacts due to transposed convolution upsampling. AEGIS performs dual-domain orthogonal decomposition:
1. **2D Fast Fourier Transform (FFT):** Computes spatial frequency components shifted to zero-frequency center: $$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j 2\pi \left(\frac{ux}{M} + \frac{vy}{N}\right)}, \quad S(u, v) = \ln(1 + |F(u, v)|)$$ Rendered with the INFERNO colormap. Unnatural symmetric starburst patterns or high-frequency spikes reveal generative grid signatures.
2. **2D Discrete Cosine Transform (DCT):** Computes real-valued harmonic energy compaction via scipy.fftpack.dct: Rendered with the VIRIDIS colormap to measure high-frequency energy ratio and spectral centroid decay.

3.4 Binary Stream Introspection & Elimination of 'N/A' Placeholders

A critical enhancement made to AEGIS resolved the common problem where social media web uploads strip camera EXIF data, causing prior forensic tools to output empty or uninformative 'N/A' tags. AEGIS performs deep binary stream introspection:
• **JPEG SOF0 Marker Parsing (0xFFC0):** Scans the bitstream for frame markers to extract component sampling ratios: Horizontal/Vertical factors (h_1, v_1) of $(2,2) \implies$ **YCbCr 4:2:0**, $(2,1) \implies$ **YCbCr 4:2:2**, $(1,1) \implies$ **YCbCr 4:4:4**.
• **JPEG DQT Luminance Quantization Table (0xFFDB):** Extracts the 64-byte quantization matrix, computes scaling ratios against standard IJG 50% tables, and calculates an estimated DCT compression quality (e.g. High Fidelity DCT (~94%)).
• **Descriptive Forensic Encodings:** Hardware fields for stripped images are intelligently populated with descriptive classifications (e.g. *'Non-Hardware (Digital Source)'*, *'Digital Raster Stream'*, *'Synthetic Sensor (Direct Pixel Matrix)'*, *'Uniform Depth of Field (Synthetic f/0)'*).

4. Court Admissibility, Legal Standards & PDF Reporting Engine

Digital forensics is only as valuable as its admissibility in a judicial forum. AEGIS Forensics adheres to the four major statutory standards:

Legal / Forensic Standard	Statutory Requirements	AEGIS System Implementation
ISO/IEC 27037:2012	Digital evidence identification, collection, acquisition, and preservation.	Auditability (case tokens), Repeatability (fixed mathematical seeds), Reproducibility (independent test suite), Justifiability (XAI exhibits).
FRE Rule 901 & 902	Authenticating evidence; self-authenticating electronic records (902(13/14)).	Dual cryptographic hashing (MD5 + SHA-256) computed at intake; EXIF capture vs. modification timeline consistency checks.
Daubert Standard (509 U.S. 579)	Scientific validity: testability, peer review, known error rate, general acceptance.	Peer-reviewed algorithms (ResNet-18, Grad-CAM, LIME, FFT), documented error rates (1.80% for text), standardized control protocols.
28 U.S.C. § 1746	Unsworn declarations under penalty of perjury.	Section 11 of generated PDF includes a formal sworn examiner declaration under penalty of perjury with digital verification seal.

4.1 Dynamic Evidence Custodian & 4-Stage Chain of Custody Ledger

In active law enforcement and enterprise IR investigations, the evidence custodian varies per case. AEGIS allows the investigator to configure the **Evidence Custodian / Lead Examiner Name** directly from the UI before executing analysis. This value is transmitted via multipart FormData, persisted in the database, and reflected throughout the **4-Stage Chain of Custody Ledger**:
• *Stage 1 — Evidence Intake & Checksum:* Bitstream received and dual-hashed (SHA-256 + MD5).
• *Stage 2 — Physical & Logical Inspection:* JPEG stream, DQT tables, and SOF0 subsampling decoded.
• *Stage 3 — Algorithmic Multi-Modal Analysis:* ResNet-18 inference, Grad-CAM heatmap, and 2D FFT/DCT executed.
• *Stage 4 — Forensic Archival & Certification:* Final findings sealed, stored in SQLite, and certified for judicial submission.

4.2 The 12-Section Court-Admissible PDF Architecture

The PDF generator in report_service.py employs a custom dynamic canvas (NumberedCanvas) using ReportLab's two-pass engine to calculate exact total page counts (Page X of Y), render running case headers (CR-FOR-2026-{id}), and running custodian footers. The generated report compiles twelve formal sections:
1. Administrative Case Information & Jurisdiction Record
2. Chain of Custody & Dual-Hash Cryptographic Verification (SHA-256 + MD5 corroboration)
3. Validated Tooling & Scientific Repeatability Registry (Daubert test compliance)
4. Physical & Logical File Specifications (SOF0 subsampling, DQT quality, bit depth)
5. Timeline & Temporal Provenance (Creation vs. modification check under FRE 901)
6. Optical Sensor & Camera Hardware Footprint (or digital synthetic designations)
7. Photographic Evidence Exhibits (Exhibit A: Original, Exhibit B: Face ROI, Exhibit C: 2D FFT, Exhibit D: Grad-CAM)
8. Multi-Modal Findings & Spectral Analysis (Spectral centroid, high-frequency energy ratio)
9. Itemized Evidentiary Findings Ledger (Structured audit trail with severity weights)
10. Forensic Expert Opinion & Certificate of Analysis (Formal judicial conclusion)
11. Statutory Examiner Attestation & Sworn Declaration (28 U.S.C. § 1746 oath with dynamic signature block)
12. Forensic Scope, Limitations & Procedural Notes (Statutory boundaries and model confidence thresholds)

5. Master Viva Voce & Technical Defense Q&A; (20 Critical Examiner Questions)

Q1: What is the primary motivation behind AEGIS Forensics?

Traditional forensic tools are fragmented, produce black-box AI predictions that fail under legal cross-examination, and lack chain-of-custody tracking. AEGIS combines email phishing detection and deepfake image forensics into a court-admissible framework complying with ISO/IEC 27037, FRE 901/902, and Daubert standards.

Q2: Why did you decouple Email and Image forensics into separate modules rather than a multi-modal fusion network?

In digital forensics jurisprudence, strict evidence separation prevents cross-contamination. Phishing text and synthetic imagery represent distinct attack vectors; a multi-modal model would create opaque dependencies where text noise could falsely bias image forgery scores. Decoupled modules guarantee independent repeatability and legal admissibility.

Q3: What machine learning model is used for Email Forensics and why?

We utilize a TF-IDF vectorizer coupled with a calibrated Logistic Regression classifier. For high-dimensional text classification, Logistic Regression achieves superior calibration, computational efficiency (~42 ms latency), robust L2 regularization against overfitting, and high accuracy (98.20% and 0.9762 F1-score) while enabling exact linear explainability via LIME.

Q4: How does LIME work mathematically in your email module?

LIME draws perturbations in the token space around an email by randomly masking words, weights these perturbations using an exponential distance kernel, and fits an interpretable sparse linear surrogate model. The learned regression coefficients represent individual word importance weights, visualized as green/red attribution bars.

Q5: What authentication headers do you inspect in email forensic analysis?

We inspect RFC 7208 (Sender Policy Framework - SPF), RFC 6376 (DomainKeys Identified Mail - DKIM), and RFC 7489 (DMARC). Additionally, we inspect RFC 5322 headers to identify discrepancies between the display 'From' address and the 'Reply-To' address, which is a classic indicator of credential harvesting.

Q6: How does the URL inspection engine detect brand spoofing?

The URL engine uses tldextract to isolate the FQDN, checks if the host is an unauthenticated raw IP, evaluates the TLD against known abuse lists (.xyz, .top), and executes Levenshtein distance calculations and character substitution checks (homograph attacks like 'paypa1.com') against a protected brand registry.

Q7: What deep learning architecture is used for Deepfake Image Forensics?

A customized ResNet-18 convolutional neural network. We leverage ResNet-18 because its spatial-residual skip connections (F(x) + x) preserve high-frequency gradient residuals that carry digital synthesis artifacts, and its lightweight footprint enables real-time CPU/GPU forensic evaluation without memory saturation.

Q8: How does Grad-CAM work mathematically, and why is Layer 4 chosen?

Grad-CAM computes the gradients of the class score with respect to feature maps in the final convolutional layer (Layer 4). By global-average-pooling these gradients and computing a ReLU-weighted linear combination, Grad-CAM captures high-level semantic areas driving the decision while filtering out negative or neutral activations.

Q9: What is the forensic significance of 2D Fast Fourier Transform (FFT) analysis?

Natural camera lenses and optical sensors produce continuous, smoothly decaying spatial frequency distributions. Generative networks (GANs/Diffusion) employ transposed convolutions and upsampling layers that create periodic grid artifacts. These appear as unnatural symmetric spikes in the log-magnitude 2D FFT spectrum.

Q10: What does the 2D Discrete Cosine Transform (DCT) reveal?

The 2D DCT transforms spatial pixel values into real-valued spatial frequency coefficients. Because DCT concentrates natural image energy into low-frequency coefficients in the top-left corner, analyzing high-frequency energy ratio and spectral centroid exposes synthetic smoothing or compression tampering.

Q11: How did you fix the problem of stripped EXIF data showing 'N/A' on web images?

We engineered low-level binary stream introspection. By scanning byte markers directly, we extract JPEG SOF0 markers to derive chroma subsampling ratios (4:2:0 vs 4:4:4) and DQT markers to estimate quantization quality (~94% DCT quality). If hardware tags are absent, the engine assigns descriptive forensic classifications rather than uninformative 'N/A's.

Q12: How is chroma subsampling calculated from the JPEG SOF0 header?

In JPEG, marker 0xFFC0 (SOF0) defines image components. By reading the sampling factors of Component 1 (Luminance Y), horizontal (h1) and vertical (v1) factors of 2 and 2 indicate 4:2:0 subsampling; 2 and 1 indicate 4:2:2; and 1 and 1 indicate 4:4:4 uncompressed chroma.

Q13: Why does AEGIS compute dual cryptographic hashes (MD5 and SHA-256)?

Under ISO/IEC 27037:2012 and FRE Rule 902(14), evidence self-authentication requires proof of bitstream integrity. SHA-256 provides modern collision resistance, while MD5 provides backwards compatibility with historical forensic databases (e.g. NIST NSRL). Dual hashing proves zero post-acquisition tampering and eliminates defense collision claims.

Q14: How does AEGIS satisfy the Daubert Standard for legal admissibility?

The Daubert test requires: 1) Empirical testability (reproducible validation datasets); 2) Peer review (ResNet, Grad-CAM, LIME, FFT are widely published); 3) Known error rates (1.80% for text, documented in Section 3 of reports); 4) Standardized controls (dual hashing, automated unit tests); 5) General scientific acceptance.

Q15: How does the Dynamic Evidence Custodian feature operate?

Investigators specify the Lead Examiner / Custodian name in the UI before running analysis. This value is transmitted via multipart FormData to FastAPI, stored in the SQLite database, and injected dynamically into the 4-stage Chain of Custody ledger, Section 11 signature block, and running PDF footers.

Q16: How does the PDF report generator calculate 'Page X of Y' dynamically?

ReportLab's SimpleDocTemplate cannot determine total pages on a single pass. We implemented a custom NumberedCanvas inheriting from canvas.Canvas that intercepts showPage(), saves page states in memory, and in a second pass (save()) renders total page counts, running headers, and footers across all pages.

Q17: What is the significance of 28 U.S.C. § 1746 in Section 11 of the PDF report?

28 U.S.C. § 1746 provides that an unsworn declaration made under penalty of perjury has the full legal force and evidentiary weight of a sworn affidavit. Section 11 binds the named examiner to the scientific veracity of the findings, rendering the document court-admissible.

Q18: What database is used, and what does the schema look like?

An ACID-compliant local SQLite database (forensics.db) using the 'analyses' table: id (primary key), module ('email'|'image'), filename, prediction, risk_level, overall_risk_score, details_json (storing full telemetry, dual hashes, custodian name, and exhibits), and timestamp.

Q19: How is the software verified for quality assurance and regression prevention?

Through an automated test suite comprising 24 unit and integration tests executed via pytest, verifying everything from cryptographic integrity and model inference to EXIF date extraction and API status codes. The frontend is validated via strict TypeScript compilation (tsc -b && vite build).

Q20: What are the primary future research directions for AEGIS Forensics?

1) Audio Deepfake Forensics using RawNet3/Spec-ResNet; 2) Video Temporal Inconsistency Forensics using 3D-CNNs or Video Vision Transformers; 3) Hardware-backed cryptographic report signing via TPM 2.0 / HSM modules.

6. Project Demonstration & Step-by-Step Execution Guide

To run AEGIS Forensics locally for evaluation or live demonstration:

Terminal 1 — Backend (FastAPI / Uvicorn on Port 8000):

```
.\venv\Scripts\Activate.ps1
python -m uvicorn backend.app.main:app --host 127.0.0.1 --port 8000 --reload
```

Terminal 2 — Frontend (Vite Dev Server on Port 5173):

```
cd frontend
npm run dev
```

Demonstration Workflow:

1. Open http://localhost:5173 in Google Chrome.
2. **Email Forensics:** Click 'Email Forensics' in the sidebar, click 'Load Sample Phishing Email', and click 'Analyze Email'. Show the evaluator the 0–100 risk score, LIME word attribution bars, URL lexical table, and RFC 5322 header parsing.
3. **Image Forensics:** Click 'Image Forensics', upload a test image, customize the **Evidence Custodian** name (e.g. *'Special Agent Jane Doe'*), and click 'Execute Image Forensic Analysis'. Demonstrate the dual MD5+SHA256 hashes, Grad-CAM activation overlay, 2D FFT spectrum, and enriched metadata.
4. **Court PDF Generation:** Click 'Export PDF Report' to open and review the 12-section court-admissible forensic certificate.